

A magnified view of stellar and AGN feedback at high redshift

Purpose. Feedback from active galactic nuclei (AGN) and stars is widely recognized as a key regulator of galaxy evolution, governing star formation, gas accretion, and the co-evolution of supermassive black holes and their host galaxies (Fabian 2012; Heckman & Thompson 2017). While AGN feedback operating through radiation, winds, and radio jets can heat, eject, or prevent the cooling of gas, stellar feedback in the form of supernovae, radiation pressure, and massive-star-driven galactic winds drives turbulence, enriches the circum- and inter-galactic medium, and regulates star-formation efficiency on local scales. These processes are particularly critical at high redshift, around Cosmic Noon at $z \sim 2$, when the cosmic star-formation rate density and black-hole accretion rate both peaked and present-day massive galaxies assembled most of their stellar mass (Madau & Dickinson 2014; Förster Schreiber & Wuyts 2020). However, even with JWST, ALMA, and the VLA, spatially resolved studies of feedback at these epochs remain extremely challenging: unlensed observations are typically limited to galaxy-integrated measurements or kiloparsec scales, blending multi-phase gas, mixed heating sources, and outflow kinematics into a single set of observables. Strong gravitational lensing offers a direct route to sub-kpc physical scales at high redshift through magnification (Treu 2010), but its full power can only be realized if differential magnification is properly accounted for and the background source is reconstructed pixel by pixel in the source plane. We therefore propose to exploit strongly lensed high-redshift galaxies with non-parametric, pixelized source-plane reconstruction techniques (Warren & Dye 2003; Suyu et al. 2006; Vegetti & Koopmans 2009; Nightingale, Dye & Massey 2018), to recover the intrinsic distribution and kinematics of star formation, molecular gas, and radio jet/outflow emission at hundreds-of-parsec resolution and to enable spatially resolved attribution of AGN and stellar feedback physics in the early Universe.

Objective 1. Hundreds-pc-scale view of AGN and stellar feedback through molecular excitation and heating in lensed high- z galaxies

Strong gravitational lensing provides the promising direct route to sub-kpc molecular-gas physics at high redshift, yet its diagnostic power remains incompletely exploited. Existing studies of lensed dusty star-forming galaxies have largely fitted galaxy-integrated CO ladders and dense-gas line ratios (e.g. Carilli & Walter 2013), implicitly assuming co-spatial emission and a single magnification factor. In practice, different transitions arise from gas phases with different spatial extents and kinematics, and therefore suffer differential magnifications (Hezaveh et al. 2012; Serjeant 2012; Spilker et al. 2015). Fitting integrated ratios thus folds differential magnification, multi-phase structure, and mixed heating sources — star formation, AGN radiation, and shocks — into one degenerate solution.

This program removes that degeneracy by moving the entire excitation analysis into the source plane, pixel by pixel, and carrying it through to a spatially resolved attribution of heating mechanisms. The outcome is a new observable class: heating maps of galaxies at Cosmic Noon and beyond, localizing where star formation, AGN radiation and mechanical input, and shocks each sustain the molecular phase. Lensed galaxies are thereby transformed from bright but blurred laboratories into spatially resolved testbeds for feedback physics, directly comparable to cosmological simulations with explicit stellar and AGN feedback channels (Springel et al. 2005; Fabian 2012; Harrison et al. 2018).

Methodology and Contents. First, a source-plane spectral data cube will be constructed. We plan to start from the PASSAGES (Planck All-Sky Survey to Analyze Gravitationally Lensed Extreme Starbursts) program archival data. Multi-transition maps — the CO ladder, isotopologues, dense-gas tracers such as HCN, HCO⁺ and HNC (Gao & Solomon 2004), and atomic carbon (Walter et al. 2011) together with dual-band dust continuum — will be reconstructed onto a dynamic-resolution, pixelized source-plane grid. Lens modelling will be performed with *PyAutoLens* (Nightingale et al. 2018). For each system a parametric mass model for the deflector and a parametric source model will first be optimized to establish a robust macro-model, including external shear and, where required, multiple lens planes. Conditioned on that best-fit macro-model, a non-parametric, pixelized source reconstruction with adaptive regularization will then be performed independently per line and per frequency channel. This two-stage procedure preserves the stability of the parametric fit while recovering irregular, clumpy intrinsic structure. It yields

both the spatial distribution and kinematics of individual gas components and a per-transition magnification map, so differential magnification is measured from the lens model rather than assumed away.

Second, physical condition maps will be derived in a pixelized source plane. Each source-plane pixel will be fitted independently with a non-LTE escape-probability solver of the RADEX class (van der Tak et al. 2007), using LTE rotation diagrams only as a baseline and cross-check. In this decomposition the CO ladder primarily constrains kinetic temperature, high-critical-density lines constrain volume density, and optically thin anchors — ^{13}CO , C^{18}O , C I and dust continuum — break the column-density degeneracy. The deliverable is a set of source-plane maps of kinetic temperature, H_2 volume density and molecular column density with full per-pixel posteriors, so that uncertainty, not only best-fit values, is spatially resolved.

Third, heating attribution maps will be performed per pixel. Measured line intensities will be fitted against photon-dominated region grids including the PDR Toolbox (Kaufman et al. 1999, 2006; Pound & Wolfire 2023), Cloudy (Ferland et al. 2017) and KOSMA-tau (Roellig et al. 2006, 2022), X-ray-dominated region grids (Meijerink & Spaans 2005; Meijerink et al. 2007), and the Paris-Durham shock grids (Flower & Pineau des Forets 2015; Godard et al. 2019; Lesaffre et al. 2013), with the RADEX-derived conditions imposed as consistency priors. This yields maps of FUV field strength, X-ray energy deposition and shock velocity and pre-shock density.

Fourth, every pixel will pass an energy-budget check in observations, since line-ratio fits alone remain degenerate. The star-formation energy budget will be constrained from resolved star-formation-rate surface density inferred from tracers like rest-frame ~ 100 micron continuum, free-free emission (Murphy et al. 2011; Kennicutt & Evans 2012) and high-lying water lines (Omont et al. 2013; Yang et al. 2013), the AGN budget from Chandra/XMM X-ray data (optional) and radio jet and outflow measurements, against the observed molecular and fine-structure cooling. A heating mechanism will be claimed only where both the heating model grid fit and the energy budget close, anchored by independent shock tracers and outflow kinematics where shocks are invoked.

Objective 2. Localized AGN-jet feedback on star formation in submillimetre galaxies (SMGs)

While galaxy-scale outflows driven by AGN winds and radiation are now routinely incorporated into galaxy-evolution models (Fabian 2012; Harrison et al. 2018), the role of small-scale, low-power jets in radio-quiet systems remains largely overlooked. Recent deep radio survey finds AGN-jet signatures in about 40% of radio-quiet quasars (Njeri et al. 2026), yet most feedback censuses still assign their radio excess to star formation or to galaxy-wide winds (Jarvis et al. 2019, 2021). Pilot work on the Cloverleaf quasar demonstrates that such localized jets do exist at $z \sim 2.6$ and can suppress star-formation efficiency by $\sim 30\%$ on the jet-hosting side of the disk through direct jet-ISM interaction on scales of hundreds of parsecs (Zhang et al. 2023; Zhang et al. submitted). Whether this is an exceptional case or a common quenching channel during the rapid-quenching phase is unknown. This program establishes, for the first time on a statistical sample, how frequently localized jets occur in SMGs and how they feedback on the star formation.

Methodology and Contents. Lens modelling and source-plane reconstruction will follow the same two-stage procedure as above: parametric lens modelling in *PyAutoLens* followed by non-parametric, pixelized source-plane reconstruction, applied independently to continuum and line data to preserve differential structure between stellar, dust, dense-gas and radio components.

First, a sample of lensed SMGs will be assembled and observed for localized-jet candidates. Star-formation rates will be estimated from ALMA rest-frame ~ 100 micron continuum and/or high-lying water emission, whose luminosity correlates tightly and slightly super-linearly with infrared luminosity (Omont et al. 2013; Yang et al. 2013), cross-checked against low-frequency VLA continuum to isolate free-free emission as a dust-unbiased star-formation tracer (Murphy et al. 2011). Systems showing radio flux in excess of that expected from star formation, or showing offset or elongated radio components, will be prioritized as candidates.

Second, confirmed candidates will be followed up at 50-500 pc effective resolution with ALMA long baselines, VLA A-configuration, and VLBI including KVN, EAVN, VLBA and other facilities, resolving jet knots, hot spots and asymmetric morphologies that are unresolved in conventional imaging.

Third, the impact on star formation will be quantified. For SMGs with confirmed localized jets, resolved star-formation-rate density maps, radio sub-structures, and molecular-gas kinematics — in particular enhanced velocity dispersion, elevated high-J to low-J CO ratios, and shock-excited emission along the jet path — will be jointly analyzed, following the methodology demonstrated in the Cloverleaf (Zhang et al. 2023; Zhang et al. submitted). Suppression or enhancement will be measured differentially with respect to asymmetric morphology, yielding a per-source estimate of jet-coupled energy versus radiative and wind budgets, and hence a re-assessment of the relative roles of multiple feedback channels and their evolution across cosmic time. Beyond its impact on star formation, jet feedback will be assessed kinematically. On galaxy scales, source-plane CO/CII cubes will be fitted with 3D tilted-ring models using *3D-Barolo* (Di Teodoro & Fraternali 2015) to separate regular disk rotation from jet-driven non-circular motions, warps and enhanced dispersion along the jet axis. The derived rotation residuals and outflow kinetic power and momentum flux will then be compared against jet power in a unified energy-budget framework to distinguish jet-driven from wind-driven outflows. In the nuclear regions, long-baseline ALMA and VLBI imaging at ~50 pc resolution will pinpoint jet-circumnuclear gas interaction sites via co-spatial radio knots and disturbed dense-gas kinematics.

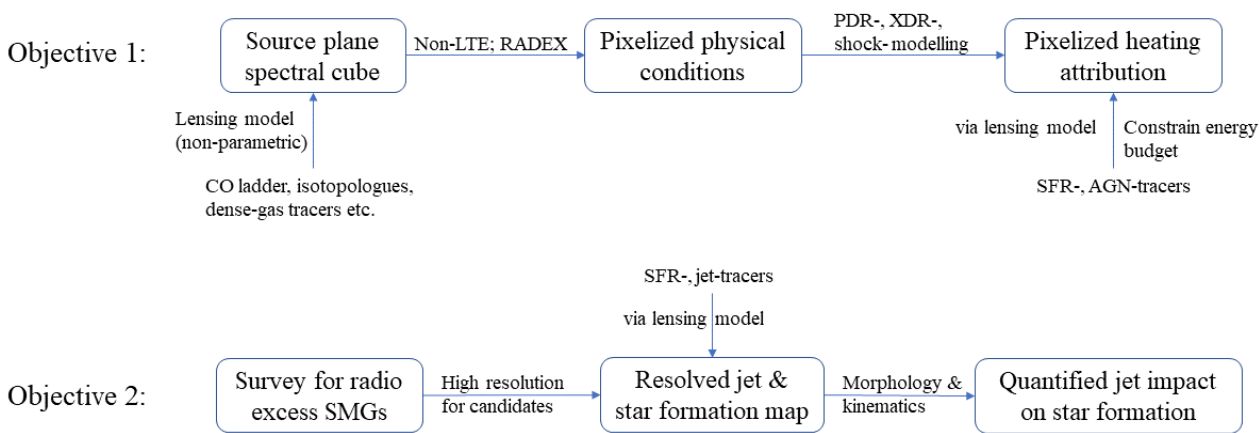


Figure 1: Workflow.

Schedule. The project is organized into two interrelated objectives (see Figure 1 for the overall workflow). The work plan is shown in the Gantt chart (Table 1; M: milestone, D: deliverable).

Objective 1: AGN and stellar feedback through molecular emissions (month 1-18). • Acquisition and reduction of archival data; source intelligence and initial parametric models; parameter optimization and non-parametric, pixelized source reconstruction (M1.1) • Pixelized physical conditions (M1.2) • Refinement of results and submission of Paper I on pixelized physical conditions (M1.3) • Refereed publication of Paper I (D1) • Heating models for pixelized heating attribution (M1.4) • Energy budget constrain in observations (M1.5) • Refinement of results and submission of Paper II on pixelized heating attribution (M1.6) • Refereed publication of Paper II (D2)

Objective 2: Localized AGN-jet feedback on star formation (month 19-36). • Observational proposals for lensed SMGs; data reduction and lensing model (M2.1) • Refinement of results and submission of Paper III on lensed SMG samples (M2.2) • Refereed publication of Paper III (D3) • Observational proposals for lensed SMGs with higher spatial resolution; pixelized source plane (M2.3) • Quantified jet feedback on star formation (M2.4) • Refinement of results and submission of Paper IV on localized jet feedback (M2.5) • Refereed publication of Paper IV (D4)

	Year 1				Year 2				Year 3			
Month	3	6	9	12	15	18	21	24	27	30	33	36
O 1	M1.1	M1.2	M1.3	M1.4	M1.5	M1.6						
				D1			D2					

O 2							M2.1	M2.2	M2.3	M2.4	M2.5	D4
									D3			

Table 1: Gantt chart of the schedule.

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